

Client non-list wire-shape audit

— ota-manager

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Anti-bluff discovery-pressure sweep (§11.4.6 no-guessing, §11.4.108 source-truth, §11.4.118 discovery-pressure) over every NON-LIST response interface in `clients/ota-manager/src/lib/api-client.ts`, following the two confirmed shape-drift bugs already fixed in this client (BUG-1 TelemetryOverview fabrication; the six list-interface `{items,total,cursor}` vs real `{items,next_cursor}` drift, commit `f711b1f9`).

Headline finding

The completeness sweep this task was designed to run **did** turn up the predicted class of defect, at a scale much larger than the conductor’s working hypothesis (“one confirmed omission — `TelemetryHistory.device_id` — plus a spot-check of the rest”). Every real, returned-by-a-handler non-list interface checked against its Go struct — 15 of them — was found to have genuine field-level drift, ranging from a single missing optional field (`TokenResponse.roles`) to entire fabricated shapes with almost no field overlap (`DeviceRegistered`, `DeviceStatus`, `DeviceHealth`, `ReleaseResponse`, `Deployment`, `RolloutState`, `RolloutDecision`, `Group`). Two of the fabricated shapes (`Deployment`, `AuditEntry`) were ALSO the per-item type nested inside an already-“fixed” list interface (`DeploymentList.items`, `AuditLogList.items`), so the earlier list-pagination fix (commit `f711b1f9`) corrected the outer envelope while the item contents underneath stayed wrong.

Per-interface audit table

Interface	Verdict	Real server citation
<code>TokenResponse</code>		<code>server/internal/api/wire.go:29-36</code>

Interface	Verdict	Real server citation
DeviceRegistered	DRIFT — missing roles?: string[]	
	DRIFT — fabricated {id, board, status, created_at}; real shape {device_id, hardware_id, device_token, token_type, expires_in, registered_at}	server/internal/api/wire.go:52-59 handlers_device.go:192-202
	DRIFT — typed as DeviceRegistered[] (the one-time registration-response shape); real per-item shape is DeviceListItem	
DeviceList.items (new DeviceListItem)	(device_id, hardware_id, model, os, os_version?, current_version?, target_version, group?, update_state, active_slot?, health_ok, last_seen?, registered_at)	server/internal/api/wire.go:81-95 handlers_device.go:167-190
DeviceStatus	DRIFT — fabricated {device_id, online, current_release_id, last_seen, ip_address}; real shape {device_id, hardware_id, current_version?, target_version, last_seen?, update_state, active_slot?, health}	server/internal/api/wire.go:69-78 handlers_device.go:204-230
	DRIFT — fabricated top-level {device_id, online,	server/internal/api/wire.go:62-65

Interface	Verdict	Real server citation
Release/ ReleaseResponse	battery_percent, storage_used_percent, temperature_celsius, uptime_seconds, last_reported}; real shape is the NESTED {ok, last_error_code} block of DeviceStatus.health — no such standalone endpoint exists DRIFT — fabricated {id, project_id, file_url, file_hash, changelog, target_board, firmware_version, created_by}; real shape {release_id, artifact_id, version, os, target_model, status, notes?, created_at}	server/internal/api/wire.go:147-1 via handlers_release.go:126-138
Deployment	DRIFT — fabricated {group_ids: string[], rollout_percentage, staged, created_by}; real shape {deployment_id, release_id, strategy, group?, status, target_count, created_at}	server/internal/api/wire.go:174-1 via handlers_deployment.go:226-237
DeploymentStatus (new DeploymentProgress)	DRIFT — fabricated {rollout_state?, failed_devices?, completed_devices?, total_devices?}; real shape is Deployment & {progress: {pending, downloading,	server/internal/api/wire.go:193-2 via handlers_deployment.go:121-132, 19

Interface	Verdict	Real server citation
	installed, succeeded, failed}} DRIFT — fabricated {progress, success_rate, error_rate, total_devices, completed_devices, failed_devices}; real shape {deployment_id, status, current_phase, phases[], updated_at}	
RolloutState (new RolloutPhaseSpec)		server/internal/api/handlers_rollout.go:16-44,67-75
RolloutDecision	DRIFT — fabricated {approved: boolean, reason}; real shape {action, reason, state: RolloutState}	server/internal/api/handlers_rollout.go:47-51,129-160
Project	DRIFT — fabricated os_types: string[] field; missing real updated_at	server/internal/api/wire.go:260-2 via handlers_project.go:95-103
Group (list item of GroupList)	DRIFT — fabricated {id, project_id, device_count, labels}; real shape {group_id, name, description?, member_count, created_at} (groups are NOT project-scoped)	server/internal/api/handlers_group.go:54-60,97-115
GroupMembers (new GroupMemberView)	DRIFT — fabricated {group_id, devices: DeviceRegistered[]}; real shape {group_id, items: {device_id, added_at}[], next_cursor} (cursor-paginated)	server/internal/api/handlers_group.go:84-95
AddGroupMembersResult	DRIFT — fabricated {added: number, failed: number}; real	server/internal/api/handlers_group.go:77-81

Interface	Verdict	Real server citation
	shape {added: string[], already_member: string[], not_found: string[]}	
Artifact	VERIFIED CORRECT — no drift	server/internal/api/wire.go:122-1
ArtifactUploadMetadata	VERIFIED CORRECT — no drift	server/internal/api/wire.go:107-1
DeltaView	VERIFIED CORRECT — no drift	server/internal/api/ handlers_delta.go:24-32
AuditEntry (new AuditActor, list item of AuditLogList)	DRIFT — actor was a bare string; real shape nests {user_id?, subject}; user_agent entirely missing	server/internal/api/audit_wire.go via :39-55
TelemetryHistory (new TelemetryEventView)	DRIFT — missing required device_id (the conductor-flagged omission); items was untyped Record<string, unknown>[] instead of the real TelemetryEventView shape	server/internal/api/ handlers_telemetry.go:15-38,66-78
TelemetryOverview	VERIFIED CORRECT (already fixed under BUG-1) — no further drift	server/internal/api/ handlers_telemetry.go:56-64
RollbackList.items (new RollbackView)	DRIFT — inline item shape {id, deployment_id, reason, created_at} dropped kind, from_release_id, to_release_id, recall_deployment_id, triggered_by, details	server/internal/api/ handlers_recall.go:23-39 via :41-4

Result: 15 of 19 checked interfaces had real field-level drift; 4 (Artifact, ArtifactUploadMetadata, DeltaView, TelemetryOverview) were already correct.

Out-of-scope discoveries (flagged, not fixed)

This task's charter is RESPONSE-shape interfaces. While reading every real handler to verify each response, the following REQUEST-body interfaces were also found drifted from the real Go request structs. They are flagged here as follow-up work items — fixing them was judged out of scope (tight-scope discipline, §11.4.20) because correcting a request body also requires correcting the create-wizard/dialog code that builds it, a materially larger change surface than this response-shape audit:

- DeviceRegistrationRequest vs real DeviceRegistration (server/internal/api/wire.go:41-49) — no field overlap at all.
- CreateReleaseRequest vs real ReleaseCreate (server/internal/api/wire.go:136-144) — no field overlap; the create-release wizard's own code comment already flags this submission path as non-functional.
- RecallRequest vs real RecallRequest (server/internal/api/handlers_recall.go:17-20) — sends {reason, force}, server requires {to_release_id, reason?}; the real endpoint would reject every recall request this client currently sends.
- CreateDeploymentRequest/CreateRolloutRequest — not cross-checked in detail (out of scope), but given the pattern above are very likely similarly drifted; noted for the same follow-up item.

Each of these has an inline comment added at its declaration site in `api-client.ts` (except the two Create* request types, noted here only, to avoid ballooning the diff) citing the real struct and file:line.

Files changed

- `clients/ota-manager/src/lib/api-client.ts` — all interface fixes listed in the table above; added DeviceListItem, DeploymentProgress, RolloutPhaseSpec, GroupMemberView, AuditActor, RollbackView, TelemetryEventView.
- `clients/ota-manager/src/types/api.ts` — re-export the new interfaces.
- `clients/ota-manager/src/hooks/useDevices.ts` — device-list adapter: DeviceListItem field remap (device_id/hardware_id/os/

current_version), new deriveViewStatus honestly derived from health_ok/update_state (previously matched a status field that never existed on the wire, so every device rendered “unknown”), targetModel now wired from the real model field (previously never populated).

- clients/ota-manager/src/features/devices/device-detail-page.tsx — DeviceStatus field remap (hardware_id/current_version/health.ok/ update_state/active_slot); currentSlot/healthStatus now wired from real fields (previously permanent placeholders); dropped the never-real ipAddress.
- clients/ota-manager/src/hooks/useDeployments.ts — d.group (real singular field) instead of d.group_ids[0] (never existed).
- clients/ota-manager/src/hooks/useDeployment.ts — dropped created_by/rollout_percentage sourcing (no real field), made both optional on the shim’s own view-model interface; targetGroupName now from d.group.
- clients/ota-manager/src/hooks/use-groups.ts — data.group_id (real primary key) instead of data.id in useCreateGroup/useUpdateGroup.
- clients/ota-manager/src/hooks/useGroups.ts — g.group_id/g.member_count instead of g.id/g.device_count.
- clients/ota-manager/src/hooks/use-releases.ts — data.release_id instead of data.id in useCreateRelease.
- clients/ota-manager/src/hooks/useReleases.ts — r.release_id/r.os/ r.target_model instead of r.id/r.firmware_version/r.target_board.
- clients/ota-manager/src/hooks/useAuditLog.ts — e.actor.subject instead of e.actor (now a nested object).
- clients/ota-manager/src/hooks/use-projects.ts — placeholder Project fixture: dropped fabricated os_types, added real updated_at.
- clients/ota-manager/src/__tests__/wire-shape-pagination.test.ts — added the now-required device_id to its pre-existing TelemetryHistory fixture (this file’s own scope is the list/pagination shape; it needed a one-field update to keep compiling under the corrected interface).
- clients/ota-manager/src/__tests__/use-telemetry-overview.test.tsx — its deploymentsFixture used the old fabricated Deployment fields (group_ids/rollout_percentage/staged); replaced with the real target_count field (the fixture’s assertions only ever depended on .items.length, unaffected).
- clients/ota-manager/src/__tests__/wire-shape-nonlist.test.ts (NEW) — the guard test for this audit; compile-time-only, tsc --

noEmit is the oracle, same pattern as wire-shape-pagination.test.ts.

No caller needed changes for: Artifact/ArtifactUploadMetadata/DeltaView (unchanged, verified correct), TelemetryOverview (unchanged, already fixed), features/audit/audit-page.tsx and features/dashboard/dashboard-page.tsx (consume the already-mapped AuditView/TelemetryOverviewView shim types, which absorbed the AuditEntry/nothing-changed-here field changes at the adapter boundary), features/deployments/deployments-page.tsx, features/deployments/ deployment-detail-page.tsx (consume the local shim Deployment view types, whose fields stayed the same — only the shim’s internal mapping changed), features/releases/releases-page.tsx, features/groups/ groups-page.tsx (same pattern — shim absorbs the wire change).

RED evidence (captured, §11.4.115)

Temporarily reverted TelemetryHistory in api-client.ts to drop the required device_id field, then ran `npx tsc --noEmit`:

```
src/__tests__/wire-shape-nonlist.test.ts(174,5): error TS2353:
Object literal may only specify known properties, and 'device_id'
does not exist in type 'TelemetryHistory'.
src/__tests__/wire-shape-nonlist.test.ts(180,3): error TS2578:
Unused '@ts-expect-error' directive.
src/__tests__/wire-shape-pagination.test.ts(60,48): error TS2353:
Object literal may only specify known properties, and 'device_id'
does not exist in type 'TelemetryHistory'.
```

Restored the fix; `npx tsc --noEmit` exit 0 (GREEN, reconfirmed below).

Additionally, during authoring, the FULL set of @ts-expect-error “old fabricated shape” assertions in wire-shape-nonlist.test.ts were verified to actually fire (not silently pass) by the intermediate tsc run that surfaced TS2578: Unused '@ts-expect-error' directive for six of them (before the object literals were reformatted to single-line so the directive aligned with the reported error line) — i.e. every “this OLD shape should be a compile error” assertion was proven to genuinely reproduce a TS error against the corrected interfaces, not merely assumed.

Verification

- `npx tsc --noEmit` — **exit 0** (GREEN, full project, all interfaces + all callers + all test fixtures).
- `npx vitest run` — **13 test files passed (13), 51 tests passed (51)**, 9.26s.

Constraints honored

- No `server/*.go` files modified (read-only, source of truth).
- No submodule, `dist/`, or other `docs/qa/` directories modified.
- No git commands run by this agent; conductor commits.